ZYNQ-FORTH

User Manual — a 16-bit Forth CPU on the Digilent Arty Z7-20

ZYNQ-FORTH
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ZYNQ-FORTH is a small interactive Forth system for the Digilent Arty Z7-20. Its CPU, the **HF16**, is a 16-bit stack machine built in the FPGA fabric (PL). Every HF16 instruction is a Forth primitive. The Zynq's ARM core loads the Forth kernel into the HF16's memory and then carries Forth's terminal over the board's USB serial port, where the **ZYNQ-FORTH app** or any terminal program talks to it.

Besides the board's LEDs, buttons and switches, Forth reaches the Arduino-compatible **shield connector**: 36 digital pins with four PWM channels, and 9 analog inputs, read by the FPGA's built-in XADC converter. On the Pmod headers JA and JB, a **pattern generator** plays 16 digital channels in steps of 5 ns, loaded from Forth or drawn in the app.

This manual covers how to use the system, how the implementation works, and every word in the kernel. For the instruction encoding and the register maps, see design.md; for the serial protocol, Protocol.md.

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1. Getting started

What you need

- A Digilent **Arty Z7-20** (the -10 has a smaller FPGA and is not supported) and a micro-USB cable to the port marked PROG/UART (J14).
- On Windows, the **ZYNQ-FORTH app**. Any other system can use a terminal program.
- To write ZYNQ-FORTH into the board's flash: the AMD/Xilinx **Vitis** or **Vivado Lab** tools (for `program_flash`), once.

Put ZYNQ-FORTH on the board

Paths in this manual are relative to the **ZYNQ-FORTH folder**: the repository, or the folder of an unpacked release. Commands run in that folder.

The board starts ZYNQ-FORTH from its QSPI flash when jumper **JP4** is on **QSPI**. To write the flash once:

1. Set JP4 to **JTAG** and switch the board on.
2. Run the flash script. It writes the boot image and verifies it: - from a release: `.\firmware\program_flash.ps1` (it writes `firmware\BOOT.bin`); - from the repository, after building (README): `.\sw\program_flash.ps1` (it writes `build\boot\BOOT.bin`).
3. Set JP4 to **QSPI** and switch the board off and on.

The script needs `program_flash` from AMD/Xilinx Vitis 2023.2 (or Vivado Lab). It finds Vitis by `XILINX_VITIS` or in `\Xilinx\Vitis\2023.2` on any drive; installed elsewhere, pass its `bin` folder: `.\firmware\program_flash.ps1 -Vitis <Vitis folder>\bin`.

LD0 lights up when Forth runs. The flash keeps ZYNQ-FORTH until something else is written there; a Forth source you save from the app (section 2.8) lives in a sector of its own.

For development, `xsct sw/run.tcl` loads the bitstream and the ARM program over JTAG instead (JP4 on JTAG; it lasts until the board is switched off).

Connect

Plug in the USB cable. Windows shows the board as a USB serial port. Its number differs from PC to PC (COM3, COM7, COM17 ...); the Device Manager lists it under *Ports*, and this manual writes it as **COMx**. Start the app, choose that port and click **Connect** (chapter 2). Or use a terminal program:

Setting	Value
Port	the board's USB serial port, COMx
Speed	115200 baud, 8 data bits, no parity, 1 stop bit, no handshake
Enter key	sends CR (PuTTY's default)
Local echo	off (Forth echoes what you type)

Only one program can use the serial port at a time. Since firmware 2.0 that program can also be a host of the HASE laboratory, which runs the board as its Forth-Lab instrument: the board notices the HASE host's protocol

and switches to it, and back as soon as a terminal program or the app sends anything. Output Forth printed meanwhile and nobody fetched then appears first (docs/Protocol.md, "HASE mode").

First session

```
[zynq-forth] 2.3, kernel 2643 words loaded and verified

HF16 Forth on Arty Z7-20
2 3 + . 5  ok
: square dup * ;  ok
7 square . 49  ok
```

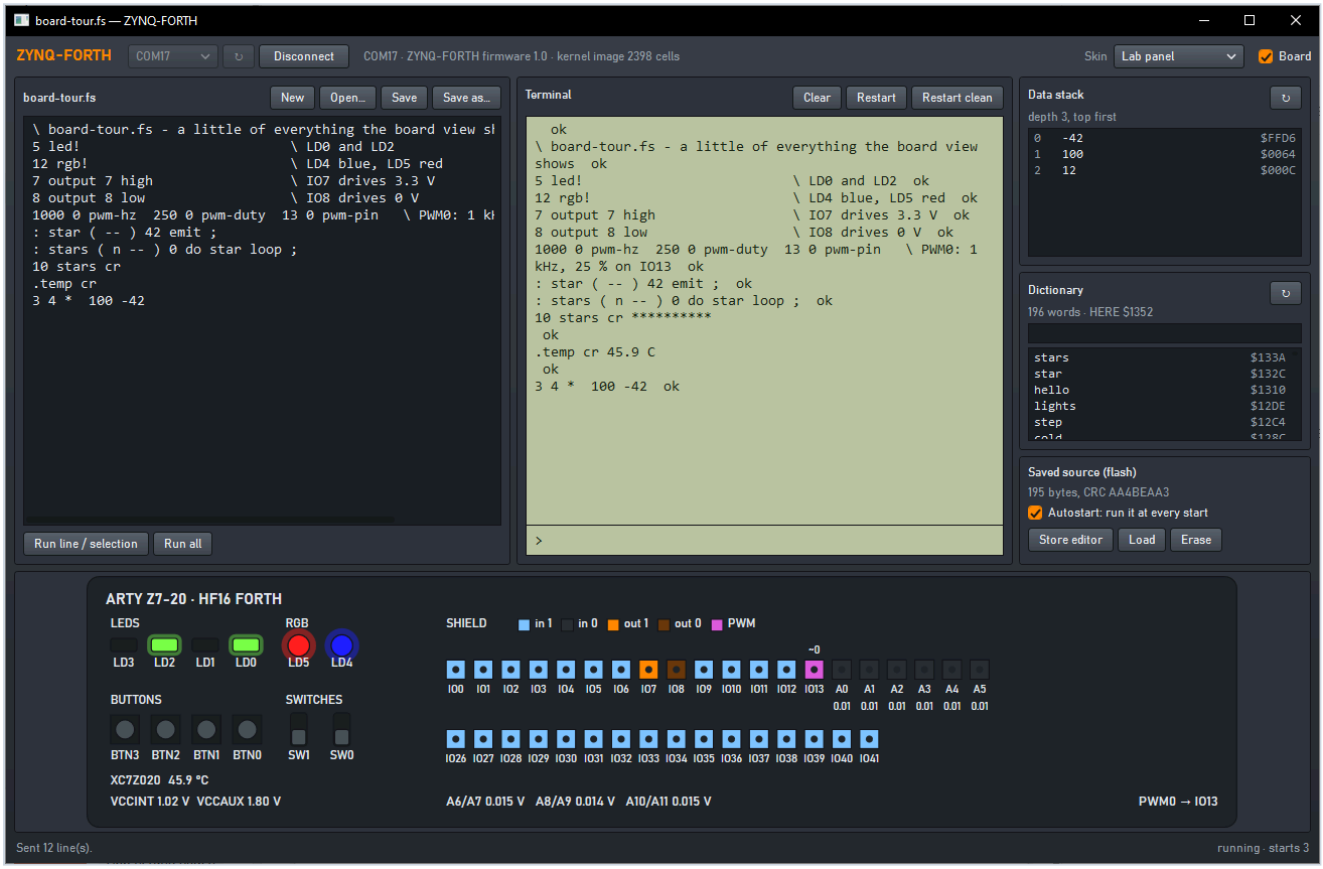
Type a line and press Enter. Forth runs the line from left to right and answers `ok`. Output appears on the same line as your input, as is usual in Forth. (To see the banner in a terminal program, connect first and then press Ctrl-R.)

Ctrl-R in a terminal program makes the ARM reload the kernel and restart Forth: the way out of an endless loop. It also removes all words you have defined. In the app, the **Restart** button does the same.

2. The ZYNQ-FORTH app

The app is a small development environment for the HF16 on Windows:

- an **editor** for Forth source files, sent to Forth in whole or in part;
- a **terminal** that sends lines to Forth and shows what it prints;
- the **data stack** and the **dictionary**, read from the running engine;
- the one **saved source** kept in the board's flash, which runs at every start;
- the **board view**: LEDs, buttons, switches, shield pins and analog inputs, live;
- five **skins**.



*The Lab panel skin after **Run all** of `examples/board-tour.fs`: LD0 and LD2 lit by `5 led!`, LD5 red and LD4 blue by `12 rgb!`, IO7 driven high, IO8 low and PWM channel 0 on IO13; the stack holds 12, 100 and -42.*

2.1 Connecting

1. Start **ZynqForth.exe**.
2. Choose the board's port in the list at the top left (↻ looks for ports again) and click **Connect**. The app remembers the port.
3. The line beside the button shows the port, the firmware version and the kernel size. The terminal shows Forth's `ok`, and the stack, dictionary and saved-source boxes fill.

If the board does not answer, the app says so and disconnects: check that ZYNQ-FORTH is on the board (chapter 1) and that no other program has the port open. **Disconnect** frees the port for a terminal program. While the app is connected, only it can use the port.

Start options, for a shortcut: `ZynqForth.exe [--skin "Lab panel"] [--connect [COMx]] [file.fs | file.pattern]` connects at once (to COMx, or without it to the port the app remembers), chooses a skin, or opens a file in the source or pattern editor.

2.2 The layout

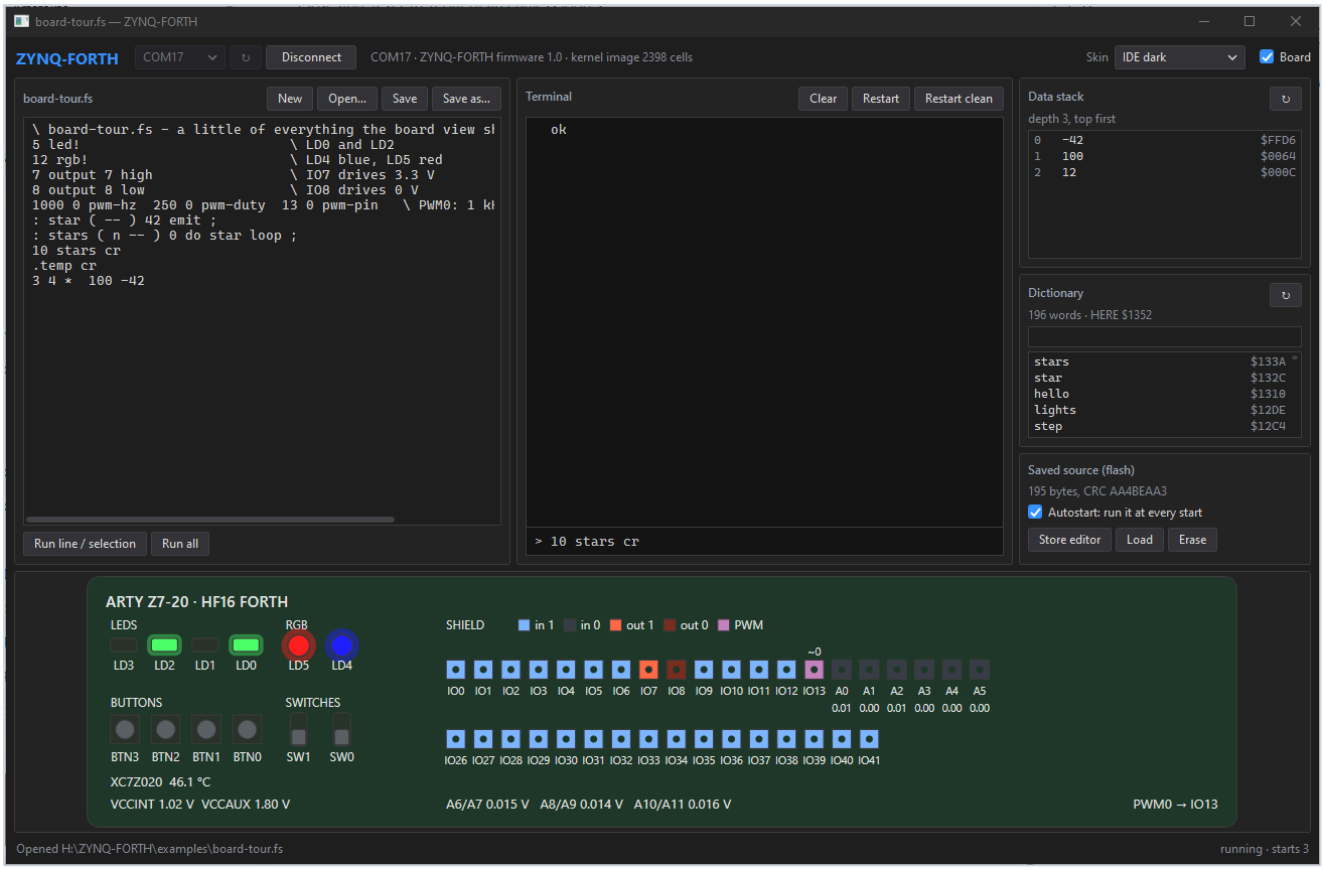
Area	Place	What it is for
Top bar	top	Port, *Connect*, firmware; **Skin**, **Board** and **Manual** (this manual as a PDF, also F1) at the right
Editor	left	*Source*: Forth source files, sent to Forth in whole or in part; *Pattern*: the pattern generator's editor (section 2.6)
Terminal	middle	Forth's output, the input line, the restarts
Data stack	top right	The stack at the last prompt, top first
Dictionary	right	The words Forth knows, newest first
Saved source (flash)	bottom right	The program in the board's flash and its autostart
Board view	below	The board's I/O, live (tick **Board**)
Status line	bottom	The last action's result (red for an error); at the right Forth's state, bytes still queued for Forth and the starts since power-up

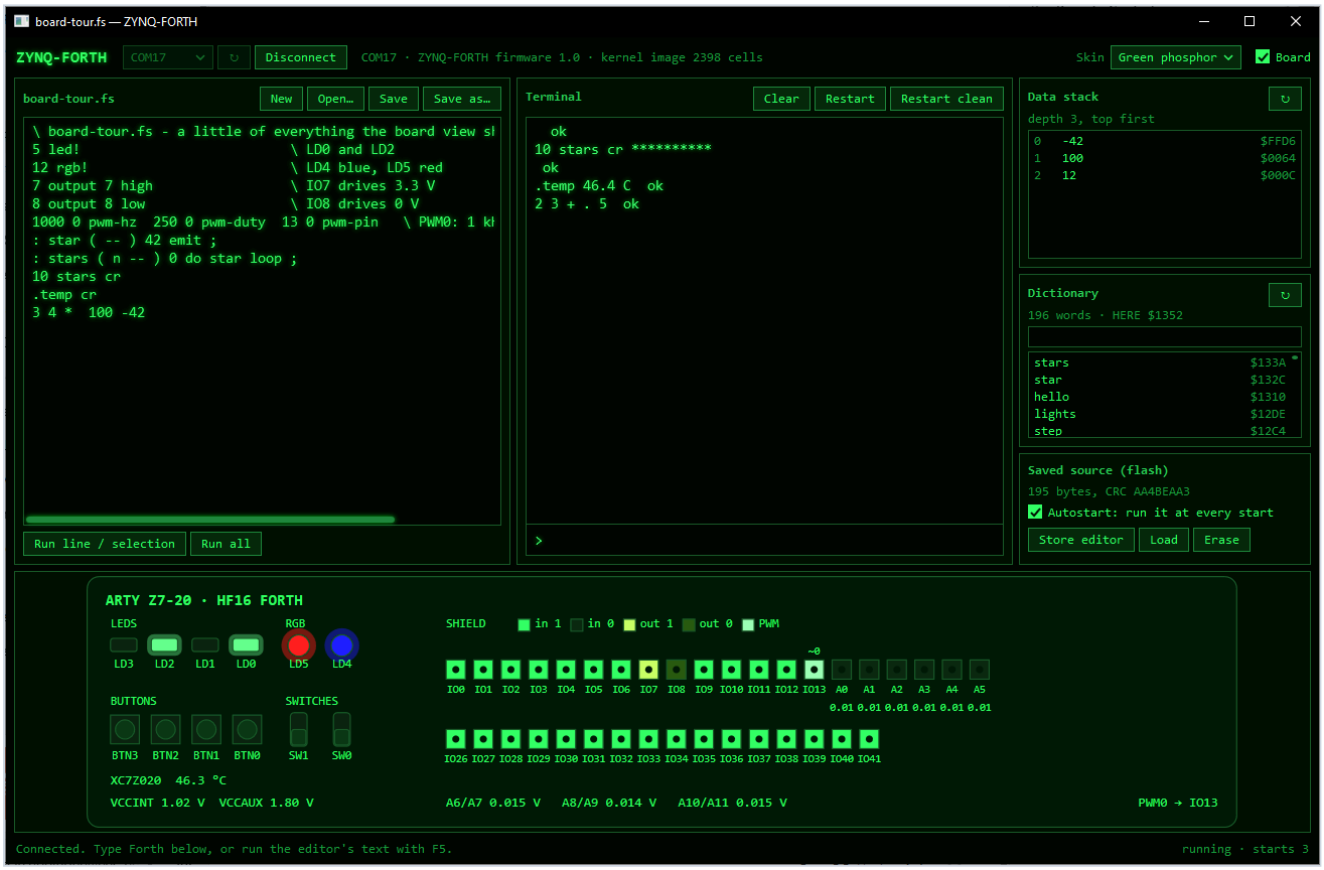
The bars between the columns, and above the board view, can be dragged.

2.3 Skins

Skin at the top right changes the look at once; the app remembers the choice.

Skin	Look
IDE dark	a dark code editor, blue accent
IDE light	the same in light colours
Green phosphor	a green CRT terminal: glowing text on black, all text monospaced
Amber	the same in amber
Lab panel	an instrument front panel with an LCD-like terminal; switches the board view on





2.4 The terminal

- **Output** shows what Forth printed since the app connected, including the echo of each line it received. **Clear** empties it; Forth is not affected.

- **Input line** (the `>` at the bottom). Type a line and press **Enter**. **Up** and **Down** bring back earlier lines. The rules of section 3.1 apply: at most 80 characters per line, plain ASCII (other characters are sent as `?`).
- **Restart** reloads the kernel and starts Forth again. All words you defined are gone, and with autostart on, the saved source runs. **Restart clean** does the same without the saved source. Restart is also the way to stop a word that never returns, such as a `begin … again` loop or one waiting for `key`.

After each line the app reads the data stack; after a line that defines words (`:`, `create`, `variable`, `constant`, …) it also reads the dictionary.

2.5 The editor

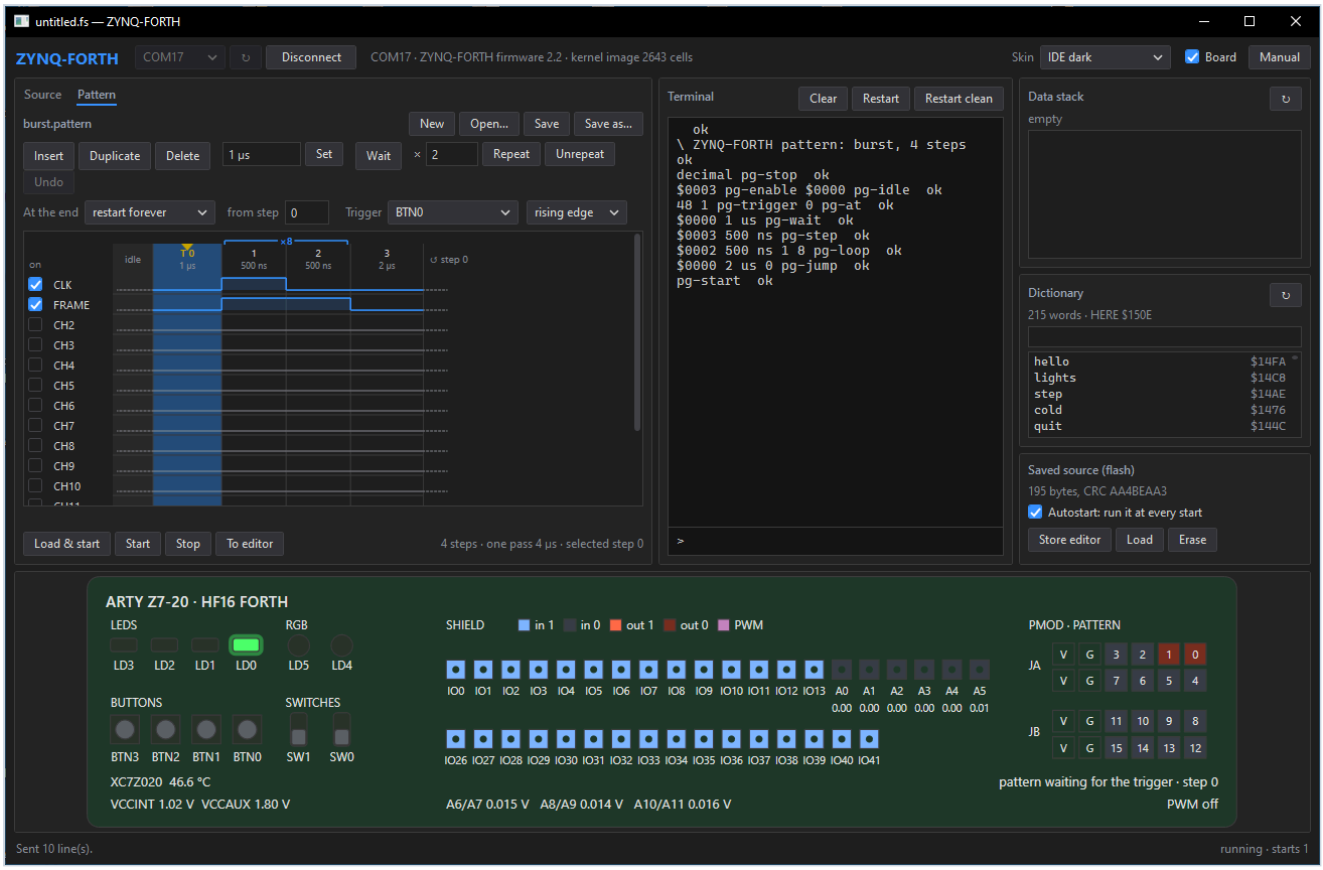
The editor area has two tabs: **Source**, the Forth editor described here, and **Pattern**, the pattern editor (section 2.6).

- The title shows the file's name, or *untitled.fs*, and a star while there are unsaved changes; the window title repeats it.
- **New**, **Open…**, **Save** (Ctrl+S) and **Save as…** work on files, usually with the extension `.fs`. *New*, *Open…*, *Load* and closing the app ask before they discard unsaved changes.
- **Run line / selection**, or **F5** in the editor, sends the selected text. With nothing selected, it sends the line with the caret and moves the caret to the next line, so F5 steps through a file line by line.
- **Run all** (Ctrl+F5) sends the whole text to Forth, as if it were typed. The status line shows the progress; long texts go in pieces as Forth reads them, so nothing is lost. When Forth has read everything, the app reads the stack and the dictionary.

The text goes to Forth exactly as typed, so the rules of chapter 3 apply. A definition may span several lines. An error discards the rest of its line only (section 3.4): the following lines still run, so check the terminal after a long run. If an error interrupted a definition, fix it and run the definition again.

2.6 The pattern editor

The **Pattern** tab above the editor turns the editor area into the pattern generator's editor (section 5.14). The column grows while it is shown; **Source** brings the Forth editor back. A pattern is a row of **steps**: each step sets the 16 channels to high or low for a duration, in ticks of 5 ns.



The timing diagram. One row per channel. The first column, **idle**, shows the idle levels: what the pins carry while the generator is stopped and before the first step (drawn dashed). Then comes one column per step. A column's header shows the step's number and duration, and long steps get wider columns, so 5 ns and one second both stay readable. Repeats are brackets above their steps, labelled ×8 or ×∞; a wait step has a **T** before its number; after the last step the diagram shows what follows: *end: idle* or ↺ *step n*. While a pattern plays on the board, a marker above the header shows the step the generator is at.

- **Click a cell** to switch the channel's level in that step; **drag** across cells to paint that level.
- **Click a cell of the idle column** to switch the channel's idle level. The edge from the idle level into step 0 is drawn like any other: it is the first edge the pin makes when the pattern starts.
- **Click a header** to select a step; **Shift**+click selects a range. The arrow keys move the selection (Shift extends it), **Del** deletes, **Ins** inserts, **Ctrl+D** duplicates, **Ctrl+Z** undoes.

Beside the diagram, each channel has its **on** box (enabled: the generator drives the pin; a channel that is not enabled is not driven) and its **name**.

Everything set here, the levels, the idle column, the trigger and what happens at the end, reaches the board only with **Load & start** (below); **Start** replays what the board already holds. The trigger only acts on **wait** steps: a pattern without one never waits.

The toolbar works on the selected steps:

Control	What it does
Insert, **Duplicate**, **Delete**	A step after the selection (like the step before it), a copy of the selection after it, or remove the selection. Repeats move with their steps.
duration box, **Set** (or Enter)	The duration of the selected steps: `100 ns`, `2.5 us`, `1 ms`, `0.5 s`; a bare number means ns. Rounded to 5 ns; 5 ns up to 21.47 s.

Control	What it does
Wait	The selected steps wait for the trigger: they show their levels, wait, then hold them for their duration.
× box, **Repeat**, **Unrepeat**	Play the selected steps this many times in all (2..65535, or *forever*); remove the repeats inside the selection. Repeats may nest four deep.
Undo	Back one change (Ctrl+Z).
At the end	*stop (idle levels)*, or *restart forever* **from step** n.
Trigger	What wait steps wait for: *software* (`pg-trig` only), BTN0..BTN3 or a shield pin, and the edge.

Checks. Under the diagram the app lists what the generator cannot do (red) and what may not work as meant (yellow), as you edit:

- errors: repeats that overlap without one containing the other, two repeats ending on the same step, more than four nested, a wait step that also ends a repeat, a restart into the middle of a repeat, a last step that ends a repeat or waits when the pattern restarts (add a step after it), more than 4096 steps;
- warnings: a repeat or restart loop whose steps are too short to keep up (section 5.14, *Timing*), wait steps with the software trigger only, and channels with high levels that are not enabled.

The buttons below:

- **Load & start** sends the pattern to the board as Forth (the terminal shows it) and starts it. It is disabled while the pattern has errors.
- **Start** (`pg-start`) plays the loaded pattern again from step 0; **Stop** (`pg-stop`) sets the idle levels.
- **To editor** puts the same Forth into the source editor (after any text there), for example to save it in flash with *Store editor* and autostart (section 2.8). The generator itself forgets the pattern at power-off.

Files. New, **Open...**, **Save** and **Save as...** work on `.pattern` files (JSON). The app also opens a `.pattern` file named on its command line. Examples are in `examples/patterns`: `counter` (a 4-bit counter on channels 0..3 in 1 µs steps), `burst` (BTN0 starts 8 clock pulses inside a frame signal) and `clock-marker` (a 1 MHz clock with a marker pulse every 1000 cycles).

2.7 The data stack and the dictionary

Both are read from the board's memory: the ARM pauses the HF16 for a few microseconds, and Forth then carries on where it was. ↻ reads again.

Data stack

- It is the stack as Forth left it at the last `ok`, exactly what `.s` would print then, but top first: before each line, `quit` copies the stack into a buffer that the ARM can read (section 4.6). While a word runs, the box keeps showing the stack of the last prompt.
- Each item shows its position (0 is the top), its value in signed decimal and in hex. The line above gives the depth, or says the stack is empty.

Dictionary

- The line above the list gives the number of words and `HERE`, the next free byte (section 4.3).
- Each word shows its flags (`immediate`, `inline`, section 4.4) and the byte address of its code in hex. Your own words come first.
- The box above the list shows only the words containing its text.

- **Double-click** a word to add it to the terminal's input line.

2.8 The saved source

The flash keeps one Forth source. With **autostart** on, it is fed to Forth after the kernel at power-up and at every restart, as if it were typed (section 4.7).

- The line under the heading shows **none**, the length in bytes and the CRC, or **damaged** (its check failed: store it again or erase it).
- **Store editor** writes the editor's text to the flash; a source already stored is replaced only after you confirm. At most 65,520 bytes. Writing takes up to about 3 seconds, and Forth's output waits meanwhile. Storing does not run the source. It keeps the autostart setting of the source it replaces; a first source is stored with autostart on. A line end is added after the last line if it has none.
- **Load** puts the saved source into the editor, for changes or a file of its own.
- **Erase** removes the saved source after you confirm.
- **Autostart: run it at every start** switches the autostart. It can be ticked only while a source is stored.

A saved source should end at the prompt. One that loops for ever, or waits for `key`, holds Forth at every start. **Hold BTN3 while switching the board on** to skip it, then switch the autostart off or erase the source. *Restart clean* skips it too.

2.9 The board view

Tick **Board** at the top right (the Lab panel skin ticks it for you). The view shows the board as Forth sees it, read four times a second:

- **LEDs** LD3..LD0 (`led!`) and the **RGB** LEDs LD5 and LD4 in their colour (`rgb!`);
- **Buttons** BTN3..BTN0 and **switches** SW1, SW0, highlighted when pressed or on;
- the **shield** pins, in the order of the board's headers: upper row IO0..IO13 and A0..A5, lower row IO26..IO41. Each pin is coloured by what it does: an input reading 1 or 0, an output driving 1 or 0, or a pin owned by a PWM channel (marked `~` and the channel number). Inputs read 1 when open because of the pull-ups (section 5.11);
- the voltages on A0..A5 under their pins, and on the differential pairs A6/A7, A8/A9, A10/A11, the chip temperature and two supplies (section 5.12);
- which PWM channels are on, and their pins;
- at the right, the Pmod headers **JA** and **JB** with the pattern generator's 16 channels, each driven high, driven low, or not enabled, and the generator's state: *stopped*, *running · step n*, *waiting for the trigger*, *ended*, and *late* when a step ran late (section 5.14). With firmware older than 2.2 the view says so instead.

The view only reads: Forth's I/O is never changed by it.

2.10 A worked session

This session writes a small program, tries it, and makes it start with the board. The program is a running light on LD0 to LD3:

```
\ lights.fs - a running light on LD0..LD3
: step ( n -- n' ) 2* dup 16 = if drop 1 then ;
: lights ( n -- ) 1 swap 0 do dup led! step 200 ms loop drop 0 led! ;
: hello ." lights ready" cr ;
hello
```

1. Click **New**, type the program into the editor, click **Save as...** and save it as `lights.fs`.
2. Click **Run all**. The terminal shows each line followed by `ok`, and the last line prints `lights ready`. `hello`, `lights` and `step` are now the first three words in the dictionary.
3. Tick **Board**. In the input line, type `10 lights` and press Enter. LD0 to LD3 light one after the other for two seconds, on the board and in the view. Press **Up** and Enter to run it again.
4. Type `1 2 3`: the data stack shows `3`, `2`, `1`, top first. Type `+` and it shows `5`, `1`. Type `drop drop`.
5. Click **Store editor**. The saved-source line shows the length and CRC, and **Autostart** is ticked.
6. Click **Restart**. Forth starts again, and the terminal shows the banner followed by the saved source and `lights ready`. From now on, `lights` is there every time the board is switched on.
7. To remove it, untick **Autostart** (the source stays in flash) or click **Erase**.

2.11 The app with a MAX1000

Since 2.4 the app also works with **MAX1000-FORTH**, the same HF16 Forth on a Trenz MAX1000 (Intel MAX 10) board, a separate project: andiepp/MAX1000-FORTH. Connect to the MAX1000's serial port as above: the app recognises the board by its `version` answer and the top bar says *MAX1000-FORTH firmware* …. The terminal, editor, data stack, dictionary and saved source work as described in this chapter. The differences:

- The board view shows the MAX1000: the 8 user LEDs, the user button, the digital pins D0..D14 and PIO1..PIO8 (Forth's pins 0–14 and 16–23), the PWM channels, and the analog inputs AIN0..AIN7 and AIN as bars with their voltage (0–3.3 V). Since 2.4.1 it also shows the chip temperature and the accelerometer's tilt in g (MAX1000-FORTH 1.0 and later).
- There is no Pattern tab: the MAX1000 has no pattern generator.
- On the MAX1000, Forth itself answers the app, not a second processor. Answers come at the prompt and while a program waits in `ms`, polls `key?`, reads `key` or prints. A word that only computes gets none until it ends, and the status line shows *no answer* meanwhile.
- **Restart** sends Ctrl-R, which the MAX1000 handles in hardware even during such a loop. *Restart clean* adds ESC, which skips the saved source. On the board, holding the user button during the start skips it too.

The MAX1000's own words (pins, ADC, temperature, accelerometer, flash) are documented in the MAX1000-FORTH manual.

3. Working with the interpreter

3.1 Input line

- A line holds up to 80 characters; anything longer is ignored.
- **Backspace** (or Delete) removes the last character.
- **Enter** runs the line.
- After the line, Forth prints `ok` if it is interpreting. While you are in the middle of a colon definition (compiling), it prints no `ok`. A definition may span several lines.

3.2 Words and numbers

The input is split at spaces. Each piece is looked up in the dictionary; if it is a word, it runs (or is compiled, inside a definition). If not, Forth tries to read it as a number. If that fails too, it is an error.

- **Names are case-insensitive:** `DUP`, `Dup` and `dup` are the same word.
- **Names** may contain any characters except space, up to 31 characters.
- **Numbers** are read in the current `base` (10 at start-up):

Input	Value
`42`	42
`-42`	−42
`$2A`	42 (hex regardless of `base`)
`-$2A`	−42
`2A` after `hex`	42

Numbers are 16 bits. Larger numbers wrap around silently: `70000 .` prints `4464` (70000 − 65536).

3.3 Stack notation

This manual describes each word's effect on the data stack as `( before -- after )`, with the top of the stack on the right. `( a b -- c )` means the word takes `b` (top) and `a`, and leaves `c`.

Symbol	Meaning
`n`	signed number (−32768..32767)
`u`	unsigned number (0..65535)
`x`	any cell
`f`	flag: 0 = false, −1 = true (any non-zero counts as true)
`c`	character or byte (0..255)
`a`	byte address
`xt`	execution token (address of a word's code)

Symbol	Meaning
`h`	header address of a dictionary entry

3.4 Errors

Message	Cause	What happens
`name ?`	word not found and not a number	data stack cleared, back to interpreting
`stack underflow`	a word took more items than were on the stack	data stack cleared
`stack overflow`	more than 20 items on the data stack	data stack cleared
`bad pin`	a pin word got a number that is not a shield pin (section 5.11)	data stack cleared
`bad channel`	`adc@` or `mv@` got a channel above 8 (section 5.12)	data stack cleared

After an error, Forth discards the rest of the line, clears the data stack and returns to interpreting. If the error happened inside a colon definition, that definition stays unfinished and hidden. It cannot be called, but it still occupies memory.

The stack check runs after each word typed at the terminal, not inside definitions. A word that pushes more than 28 items wraps the stack pointer around, and this is reported as underflow.

4. How the implementation works

4.1 The pieces

```
app / PuTTY ── USB-UART ── ARM ──── AXI ── HF16 system in the FPGA
                                          ├─ HF16 CPU, 50 MHz
                                          ├─ 16 KB RAM: kernel + your words
                                          ├─ 32-byte receive and send FIFOs
                                          ├─ LEDs, RGB LEDs, buttons, switches, ms timer
                                          ├─ shield pins: 36 digital inputs/outputs
                                          ├─ XADC: 9 analog inputs, temperature, supplies
                                          └─ pattern generator: 16 Pmod channels, 5 ns steps
```

Part	File
HF16 CPU	`rtl/hf16_cpu.v`
RAM, FIFOs, I/O, shield pins, AXI interface	`rtl/forth_system.v`, `rtl/fifo.v`
XADC and its register scanner	`rtl/xadc_scan.v`
Pattern generator, on its own 200 MHz clock	`rtl/patgen.v`
Forth kernel, written in Forth	`forth/kernel.fs`
Cross-compiler and simulator	`tools/hf16.py`
ARM loader, terminal and service requests	`sw/src/main.c`, `sw/src/flash.c`
The app	`app/src`

4.2 The HF16 CPU

- **16-bit cells.** Numbers, addresses and flags are all 16 bits.
- **Two stacks in hardware**, 32 entries each. The **data stack** holds parameters. Its top (T) is a register, and the item below it (N) is directly available, so `+`, `swap` or `over` each take one clock. The **return stack** holds return addresses, loop indices and items moved there with `>r`.
- **One instruction per clock** at 50 MHz, so about 50 million primitives per second.
- **Five instruction types:** literal (push a number 0..32767), jump, conditional jump, call, and ALU. A single ALU instruction can also move the stack pointers, store to memory and return from a subroutine.
- **Fast calls and returns:** a call takes one clock. A return is usually merged into the last instruction of a word for free.

4.3 Memory map

Address	Content
`$0000`	jump to `cold` (start-up)
`$0002`–`$14A5`	the kernel: 212 words with their headers, variables and buffers
`$14A6`–`$3FFF`	free for your definitions (about 10.8 KB). `here` shows the next free byte

Address	Content
`$4000`–`$7FFF`	a mirror of `$0000`–`$3FFF`
`$8000`–`$80D5`	I/O registers (see below)

Cells are stored little-endian, low byte first. Cell addresses must be even: `@` and `!` ignore bit 0. There is no check for a full dictionary. Writing past `$3FFF` wraps around and overwrites the kernel.

The I/O registers can also be read and written directly with `@` and `!`:

Address	Read	Write
`$8000`	next received character (removes it from the receive FIFO)	send a character
`$8002`	bit 0 = a character has arrived, bit 1 = room to send	–
`$8004`	LD0..LD3	LD0..LD3
`$8006`	bits 0..3 = BTN0..BTN3, bits 4..5 = SW0..SW1	–
`$8008`	RGB LEDs, see `rgb!`	RGB LEDs
`$800A`	millisecond counter	–
`$8010`, `$8018`, `$8020`	shield port 0, 1, 2: pin levels	–
`$8012`, `$801A`, `$8022`	port 0, 1, 2: output latch	output latch
`$8014`, `$801C`, `$8024`	port 0, 1, 2: direction, 1 = output	direction
`$8040`–`$807E`	XADC registers 0..31 (`xadc@`)	–
`$8080`–`$809E`	PWM channels 0..3 (section 5.13)	PWM registers
`$80C0`–`$80D4`	pattern generator (section 5.14)	pattern generator

Section 5.11 explains the ports, 5.12 the XADC registers, 5.13 the PWM registers and 5.14 the pattern generator's.

4.4 Dictionary entries

Every word has a header followed by its code:

```
h:      link     cell: address of the previous header (0 ends the list)
h+2:    count    byte: bits 0..4 = name length, bit 5 = inline, bit 6 = hidden,
                       bit 7 = immediate
h+3:    name     the name's characters, padded to an even address
xt:     code     HF16 instructions
```

`latest` holds the address of the newest header. Searching starts there and follows the links backwards, so a new definition with an existing name replaces the old one for everything compiled afterwards.

- **Immediate** words run even while compiling. Examples: `if`, `;`, `."`.
- **Hidden** marks a word whose definition isn't finished yet, so it can't be found.
- **Inline** words are copied into new definitions instead of being called. The primitives `dup`, `+`, `@` and others are single instructions. The return-stack words `>r`, `r>`, `r@`, `i` and `unloop` *must* be inline: as subroutines they would disturb their own return address.

4.5 What a colon definition compiles to

`: name ... ;` lays down a header and then, word by word:

In the source	Compiled as
a primitive (`dup`, `+`, `!`, `>r`, ...)	its instruction(s), copied inline
any other word	one call instruction
a number 0..32767	one literal instruction
any other number	a literal and an `invert` instruction
`if` / `while` / `until`	a conditional jump
`else` / `again` / `repeat`	a jump
`do`	`swap >r >r`, inline
`loop` / `+loop`	a call to `(loop)` / `(+loop)`, a conditional jump back, then `unloop`
`." text"`	a call to `(.")` followed by the text
`;`	a return instruction

Example: `: square dup * ;` compiles to three instructions: `dup`, `*`, return.

The cross-compiler that builds the kernel also merges a word's final return into the last instruction when that is free, and turns a final call into a jump. Definitions typed at the terminal don't get these two optimizations.

4.6 Start-up

After the ARM releases the HF16 from reset, the CPU starts at address 0, which jumps to `cold`. `cold` sets `base` to 10, prints the banner, switches LD0 on and enters `quit`, the interpreter loop. `quit` copies the data stack for the app's stack view (`snap`), reads a line (`accept-line`), interprets it (`interpret`), prints `ok`, and repeats.

4.7 The saved program (autostart)

One Forth source can be saved in the flash (from the app, section 2.8, or with `python tools/zforth.py store:<file>:on`). With autostart on, the ARM feeds it to Forth right after `cold`, at power-up and at every restart, exactly as if it were typed: the definitions are compiled and the commands run. The kernel itself is never changed, so a faulty source cannot break the start for good:

- **Hold BTN3 while switching the board on** (until the prompt appears, about a second): the saved source is skipped and Forth starts with the bare kernel. The same holds at a restart or Ctrl-R; the app also offers *Restart clean*, which skips it.
- A source that never returns to the prompt (`begin again`, waiting for `key`) blocks all further input; restart without autostart, then fix or erase the saved source.

At most 65,520 bytes can be saved; lines end in CR. While the flash is written (up to about 3 s) Forth's output waits.

5. Word reference

The kernel has 212 words. Section 5.21 lists the internal ones, which mainly serve the interpreter and compiler. Where a word differs from standard Forth, its entry says so.

5.1 Data stack

Word	Stack	Description
`dup`	`( x -- x x )`	Copy the top item.
`drop`	`( x -- )`	Remove the top item.
`swap`	`( x1 x2 -- x2 x1 )`	Exchange the top two items.
`over`	`( x1 x2 -- x1 x2 x1 )`	Copy the second item to the top.
`nip`	`( x1 x2 -- x2 )`	Remove the second item.
`tuck`	`( x1 x2 -- x2 x1 x2 )`	Copy the top item below the second.
`rot`	`( x1 x2 x3 -- x2 x3 x1 )`	Rotate the third item to the top.
`-rot`	`( x1 x2 x3 -- x3 x1 x2 )`	Rotate the top item to third place.
`2dup`	`( x1 x2 -- x1 x2 x1 x2 )`	Copy the top two items.
`2drop`	`( x1 x2 -- )`	Remove the top two items.
`?dup`	`( x -- x x | 0 )`	Copy the top item only if it is not zero.
`depth`	`( -- n )`	Number of items on the data stack, before `depth` itself pushes.
`.s`	`( -- )`	Show the stack without changing it: `<3> 1 2 3`, with the depth and the top on the right.

```
1 2 3 rot .s <3> 2 3 1  ok
```

5.2 Return stack

The return stack holds return addresses and, inside `do` loops, the loop index and limit. You can use it for temporary storage within one definition, but every `>r` needs a matching `r>` before the word ends or leaves a loop.

Word	Stack	Description
`>r`	`( x -- ) R: ( -- x )`	Move the top item to the return stack.
`r>`	`( -- x ) R: ( x -- )`	Move the top of the return stack back.
`r@`	`( -- x ) R: ( x -- x )`	Copy the top of the return stack.

These three are inline. They work inside definitions; typed directly at the terminal they would corrupt the interpreter's own return address.

5.3 Arithmetic

All arithmetic is on 16 bits and wraps around without warning.

Word	Stack	Description
`+`	`( n1 n2 -- n3 )`	Sum.
`-`	`( n1 n2 -- n3 )`	Difference n1 − n2.
`*`	`( n1 n2 -- n3 )`	Product, low 16 bits; one instruction (hardware multiplier).
`/`	`( n1 n2 -- n3 )`	Quotient, rounded toward zero: `-7 2 /` gives −3.
`mod`	`( n1 n2 -- n3 )`	Remainder, with the sign of n1: `-7 2 mod` gives −1.
`/mod`	`( n1 n2 -- rem quot )`	Remainder and quotient (symmetric division).
`u/mod`	`( u1 u2 -- rem quot )`	Unsigned division: `-7 2 u/mod` gives 1 32764.
`*/`	`( n1 n2 n3 -- n4 )`	n1 × n2 / n3 with a 32-bit intermediate product, so the product may exceed 16 bits: `3000 1000 4096 */` gives 732. Rounds toward zero.
`um*`	`( u1 u2 -- ulo uhi )`	Full 32-bit unsigned product as two cells, the high cell on top: `-1 -1 um*` gives 1 −2 (65534).
`um/mod`	`( ulo uhi u -- rem quot )`	Divide a 32-bit unsigned number by u. The quotient must fit in 16 bits (uhi < u).
`1+`	`( n -- n+1 )`	Add one.
`1-`	`( n -- n-1 )`	Subtract one.
`2*`	`( n -- n*2 )`	Double.
`2/`	`( n -- n/2 )`	Arithmetic shift right; keeps the sign (`-8 2/` gives −4).
`negate`	`( n -- -n )`	Change the sign.
`abs`	`( n -- u )`	Absolute value. `-32768 abs` stays −32768 (as u: 32768).
`min`	`( n1 n2 -- n )`	Smaller of two signed numbers.
`max`	`( n1 n2 -- n )`	Larger of two signed numbers.

Division is done in software by shift-and-subtract (16 steps). Division by zero gives no error: `u/mod` then returns −1 as the quotient. `*/` and `um/mod` give a meaningless result if the quotient doesn't fit in 16 bits.

`*/` is the usual way to scale with fractions: `x 3320 4096 */` multiplies x by 3320/4096 without losing the high bits of the product. `um*` gets its high cell from a dedicated CPU instruction (the high half of the hardware multiplier).

5.4 Logic and shifts

Word	Stack	Description
`and`	`( x1 x2 -- x3 )`	Bitwise AND.
`or`	`( x1 x2 -- x3 )`	Bitwise OR.
`xor`	`( x1 x2 -- x3 )`	Bitwise exclusive OR.
`invert`	`( x -- ~x )`	Invert all bits. Also turns a true flag into false and back.
`lshift`	`( x u -- x' )`	Shift left by u bits; u ≥ 16 gives 0.
`rshift`	`( x u -- x' )`	Logical shift right by u bits (fills with 0); u ≥ 16 gives 0.

5.5 Comparison

The comparison words return −1 (true) or 0 (false). `if`, `until` and `while` treat any non-zero value as true.

Word	Stack	Description
`=`	`( x1 x2 -- f )`	Equal.
`<>`	`( x1 x2 -- f )`	Not equal.
`<`	`( n1 n2 -- f )`	n1 less than n2, signed.
`>`	`( n1 n2 -- f )`	n1 greater than n2, signed.
`u<`	`( u1 u2 -- f )`	Less than, unsigned: `-1 1 u<` is false.
`u>`	`( u1 u2 -- f )`	Greater than, unsigned.
`0=`	`( x -- f )`	x is zero; also a logical NOT.
`0<>`	`( x -- f )`	x is not zero.
`0<`	`( n -- f )`	n is negative.
`0>`	`( n -- f )`	n is positive.
`within`	`( n lo hi -- f )`	lo ≤ n < hi; works for signed and unsigned ranges.

5.6 Memory

Word	Stack	Description
`@`	`( a -- x )`	Fetch the cell at a (bit 0 of a ignored).
`!`	`( x a -- )`	Store x at a.
`c@`	`( a -- c )`	Fetch the byte at a.
`c!`	`( c a -- )`	Store the low byte of c at a.
`+!`	`( n a -- )`	Add n to the cell at a.
`?`	`( a -- )`	Print the cell at a, like `@ .`.
`cells`	`( n -- n*2 )`	Size of n cells in bytes.

Word	Stack	Description
`cell+`	`( a -- a+2 )`	Address of the next cell.
`aligned`	`( a -- a' )`	Round a up to an even address.
`cmove`	`( a1 a2 u -- )`	Copy u bytes from a1 to a2, lowest address first.

```
variable x  5 x !  3 x +!  x ? 8  ok
```

5.7 Dictionary space

New definitions and data are added at `here`, the next free byte.

Word	Stack	Description
`here`	`( -- a )`	Address of the next free byte.
`allot`	`( n -- )`	Reserve n bytes at `here` (a negative n gives memory back).
`align`	`( -- )`	Make `here` even.
`,`	`( x -- )`	Align, store x at `here`, advance by one cell.
`c,`	`( c -- )`	Store a byte at `here`, advance by one byte.

```
create table 1 , 2 , 4 , 8 ,
table 2 cells + ? 4  ok
```

5.8 Terminal input and output

Word	Stack	Description
`key`	`( -- c )`	Wait for a character from the terminal.
`key?`	`( -- f )`	True if a character is waiting (does not take it).
`emit`	`( c -- )`	Send a character (waits while the send FIFO is full).
`cr`	`( -- )`	New line (sends CR and LF).
`space`	`( -- )`	Send one space.
`spaces`	`( n -- )`	Send n spaces (nothing for n ≤ 0).
`type`	`( a u -- )`	Send u characters starting at a.
`.`	`( n -- )`	Print a signed number in the current base, then a space.
`u.`	`( u -- )`	Print an unsigned number, then a space: `-1 u.` gives 65535.
`." text"`	`( -- )`	Compile only: print `text` when the definition runs.
`.( text)`	`( -- )`	Print `text` immediately, in a definition or at the terminal.

Digits above 9 print as capital letters: `255 hex . decimal` gives `FF`. (In `hex 255 .` the number is already read in base 16, so it prints `255`.)

```
: hello ." Hello, Arty!" cr ;
hello Hello, Arty!
 ok
```

5.9 Number base

Word	Stack	Description
`base`	`( -- a )`	Variable holding the base for number input and output (2..36).
`hex`	`( -- )`	Set the base to 16.
`decimal`	`( -- )`	Set the base to 10.

`5 2 base ! . decimal` prints 5 in binary: `101`. The base applies to input as well, so the 5 has to be typed before switching. After `2 base !`, typing `5` gives `5 ?`. An error doesn't reset `base`; type `decimal` to get back.

5.10 Board I/O

Word	Stack	Description
`led!`	`( n -- )`	Set LD0..LD3 from bits 0..3 of n (1 = on).
`led@`	`( -- n )`	Read back the LED state.
`btn@`	`( -- n )`	Buttons: bit 0 = BTN0 … bit 3 = BTN3, 1 = pressed. Debounced (sampled every 10 ms).
`sw@`	`( -- n )`	Switches: bit 0 = SW0, bit 1 = SW1, 1 = on.
`rgb!`	`( n -- )`	RGB LEDs: bits 0/1/2 = LD4 red/green/blue, bits 3/4/5 = LD5 red/green/blue.
`rgb@`	`( -- n )`	Read back the RGB LED state.
`ms`	`( n -- )`	Wait n milliseconds (0..32767).
`ms@`	`( -- u )`	Millisecond counter; wraps every 65.536 s.

The RGB LEDs run at full brightness, which is very bright; there is no dimming.

```
5 led!            \ LD0 and LD2 on
btn@ .            \ hold BTN1 while pressing Enter: prints 2
9 rgb!            \ LD4 red, LD5 red
```

5.11 Shield digital pins

The shield connector's digital pins are numbered like their labels on the board:

Pin numbers	Shield label	Pull-up
0..13	IO0..IO13 (outer digital header)	yes
14..19	A0..A5, the analog pins used as digital pins	no
26..41	IO26..IO41 (inner digital header)	yes

Every pin is an **input after start-up** and after Ctrl-R. `output` makes it drive the level of its output latch, which starts low.

Word	Stack	Description
`output`	`( pin -- )`	Make the pin an output.
`input`	`( pin -- )`	Make the pin an input again.
`high`	`( pin -- )`	Set the pin's output latch to 1 (3.3 V when it is an output).
`low`	`( pin -- )`	Set the output latch to 0.
`toggle`	`( pin -- )`	Invert the output latch.
`pin!`	`( f pin -- )`	`high` if f is true, otherwise `low`.
`pin@`	`( pin -- f )`	Read the level at the pin: true = high. Also works on outputs, where it reads back the driven level.
`pin`	`( n -- mask a )`	Bit mask and port address of pin n; `bad pin` for numbers that are not shield pins.
`bset`	`( mask a -- )`	Set the mask bits in the cell at a (a general helper).
`bclr`	`( mask a -- )`	Clear the mask bits in the cell at a.

`high` and `low` work on inputs too: they only set the latch. Setting the level first and then calling `output` makes the pin start at that level without a short glitch.

```
13 output  13 high          \ IO13 to 3.3 V
13 pin@ . -1  ok
13 low  13 input            \ back to an input
2 pin@ . -1  ok             \ an open input with pull-up reads high
```

Electrical. The pins are **3.3 V only; they are not 5 V tolerant.** On the board, each pin has a 200 Ω series resistor. Keep the current to a few mA, which is enough for an LED with a resistor. IO0..IO13 and IO26..IO41 have weak pull-ups (tens of kΩ). An open input reads high, and a button between the pin and GND needs no resistor. A0..A5 have no pull-up, because a pull-up would shift the analog readings on the same pins. As inputs, their divider to GND pulls them low.

Speed. `toggle`, `high`, `low` and `pin@` take about 1 µs each, so a loop of `toggle` makes a square wave of a few hundred kHz. Each input is synchronized to the 50 MHz clock through two flip-flops.

Whole ports. The pins are bits of three 16-bit ports. Pin n is bit n mod 16 of port n / 16. Each port has three registers: pin levels at `$8010 + 8p`, the output latch at `$8012 + 8p`, and the direction at `$8014 + 8p` (1 = output). Writing whole ports changes several pins at once:

```
255 $8014 !                 \ IO0..IO7 as outputs
170 $8012 !                 \ IO1, IO3, IO5, IO7 high, the others low
$8010 @ 255 and .           \ read IO0..IO7 at once
```

5.12 Analog inputs

The Zynq contains the **XADC**, a 12-bit analog-to-digital converter. It measures nine shield inputs, the chip temperature and six supply voltages. It runs continuously in the background: each value is an average of 16

conversions and is refreshed about every millisecond. Reading a value takes no waiting, but after the input has changed, the reading needs a few milliseconds to settle: wait `10 ms` before reading.

Channel	Shield pins	Input range	`mv@` returns
0..5	A0..A5	0..3.3 V to GND	0..3319 mV
6	A6 (+) and A7 (−)	0..1 V, A6 above A7	0..999 mV
7	A8 (+) and A9 (−)	0..1 V	0..999 mV
8	A10 (+) and A11 (−)	0..1 V	0..999 mV

Word	Stack	Description
`adc@`	`( ch -- u )`	Raw result of channel 0..8: 0..4095 for 0 to full scale. `bad channel` for other numbers.
`mv@`	`( ch -- mv )`	The channel's voltage in millivolts at the shield pins.
`temp@`	`( -- n )`	Chip temperature in tenths of a degree Celsius: 452 means 45.2 °C.
`.temp`	`( -- )`	Print the chip temperature, for example `45.2 C`.
`supply@`	`( reg -- mv )`	A supply voltage in millivolts, from XADC register reg (table below).
`xadc@`	`( reg -- u )`	Raw 12-bit result of XADC status register reg (0..31).
`vaux`	`( -- a )`	Table of 9 bytes: the XADC auxiliary input of each channel (used by `adc@`).

reg	Supply	Nominal
1	VCCINT, FPGA logic core	1000 mV
2	VCCAUX, FPGA auxiliary	1800 mV
6	VCCBRAM, block RAM	1000 mV
13	VCCPINT, ARM core	1000 mV
14	VCCPAUX, ARM auxiliary	1800 mV
15	VCCO_DDR, DDR3 memory interface	1500 mV

```
.temp 44.7 C  ok
0 mv@ . 5  ok                \ A0 open
1 supply@ . 2 supply@ . 15 supply@ . 1018 1801 1489  ok
```

The readings shown in this section come from an Arty Z7-20 at room temperature. Your values will differ slightly.

A0..A5 pass through a 2.32 kΩ / 1 kΩ divider on the board, so 3.32 V at the pin gives the converter's full 1 V. Voltages above 3.3 V read as 3319 mV. The divider loads the signal with about 3.3 kΩ to GND, so a source with a high impedance reads too low.

A6..A11 form three pairs and go to the converter without a divider. Each channel measures the + pin against the − pin, from 0 to 1 V. For a single-ended signal, connect the − pin to GND. Keep both pins between 0 and 1 V; a negative difference reads 0.

The A0..A5 pins are also digital pins 14..19 (section 5.11). Using one as an output drives the analog input of the same pin, which is handy as a self-test: `14 output 14 high 10 ms 0 mv@ .` prints about 3250 on A0. Make the

pin an input again (`14 input`) before you measure an external signal on it.

Accuracy. The converter is calibrated automatically. The divider's 1 % resistors and the 200 Ω series resistors add an error of a few percent on A0..A5. An open A0..A5 input reads a few mV, an open A6..A11 pair about 15 mV. The temperature is the die temperature, typically 10–20 °C above the room.

5.13 PWM

Four PWM channels, 0..3, make square waves of variable frequency and duty cycle in the FPGA, independently of Forth: once set, a channel runs on its own, also while Forth computes or waits for input. Each channel drives **any shield digital pin** (section 5.11).

Word	Stack	Description
`pwm-pin`	`( pin ch -- )`	Route channel ch to the pin and start it. The pin becomes an output driven by the channel; `bad pin` for numbers that are not shield pins.
`pwm-off`	`( ch -- )`	Stop the channel. Its pin returns to its own latch and direction (an input after start-up).
`pwm-hz`	`( u ch -- )`	Frequency in Hz, 1..65535.
`pwm-khz`	`( u ch -- )`	Frequency in kHz, 1..25000.
`pwm-duty`	`( permille ch -- )`	Duty cycle in ‰ (tenths of a percent), 0..1000: 0 is always low, 1000 always high; values outside are clamped.
`pwm-clocks`	`( ud ch -- )`	The period directly, in 50 MHz clocks (20 ns), as a double number: low cell, then high cell. At least 2.
`pwm`	`( ch -- a )`	Register address of channel ch; `bad channel` for ch above 3.
`ud/u`	`( ud u -- ud' )`	Unsigned double divided by an unsigned single, giving a double quotient (used by `pwm-hz`).

The order does not matter: frequency, duty and pin can be set in any order and changed at any time. **A change takes effect at the end of the running cycle,** so a pulse is never cut short and the frequency or duty can be swept while the channel runs. After start-up, and after a restart, every channel is off with frequency 25 MHz and duty 0.

```
1000 0 pwm-hz  250 0 pwm-duty  13 0 pwm-pin   \ IO13: 1 kHz, 25 % high
500 0 pwm-duty                                \ now 50 %
2 1 pwm-hz  100 1 pwm-duty  12 1 pwm-pin      \ IO12: 2 Hz, 50 ms pulses
0 pwm-off  1 pwm-off
```

Resolution. The period is counted in clocks of 20 ns, so a frequency is exact when 50,000,000 divided by it is a whole number; otherwise the period is rounded down. The duty cycle is set in steps of 1/65536 of the period, and the pulse is a whole number of clocks: at 1 kHz (50,000 clocks) the duty can be set to about 0.002 %, at 1 MHz (50 clocks) only in steps of 2 %. For a fine duty cycle, keep the frequency well below 1 MHz.

Two channels on the same pin are combined (OR). The pin's level can be read with `pin@` like any output; a loop of `pin@` samples the waveform about every 1.5 µs.

Checking it without an instrument. A0 is pin 14 and also analog channel 0, so a PWM channel on pin 14 can be read back as a voltage: the converter's averaging shows the duty cycle as a share of the 3.3 V level, with some scatter at high frequencies.

```
100 2 pwm-khz  14 2 pwm-pin
250 2 pwm-duty  20 ms  0 mv@ . 880  ok
750 2 pwm-duty  20 ms  0 mv@ . 2413  ok
1000 2 pwm-duty 20 ms  0 mv@ . 3250  ok
2 pwm-off
```

Registers. Channel c is at `$8080` + 8c:

Address	Register
+0	period, low 16 bits (taken when the high half is written)
+2	period, high 16 bits; writing it sets the period in clocks
+4	duty, in 1/65536 of the period; `$FFFF` = always high
+6	bit 15 = on, bits 5..0 = pin number

5.14 Pattern generator

The pattern generator plays up to 4096 **steps** on **16 digital channels** in ticks of **5 ns**, from its own 200 MHz clock. A step sets all 16 levels for a duration of 1 tick (5 ns) up to 4,294,967,295 ticks (21.47 s) and says what happens after it: the next step, a repeat, a jump, a wait for a trigger, or the end. Forth loads the steps and starts it; from then on the generator runs on its own, also while Forth computes or waits for input. The app draws patterns as a timing diagram and loads them for you (section 2.6).

Pinout. The channels are on the two Pmod headers **JA** (channels 0..7) and **JB** (channels 8..15). A Pmod header has two rows of six pins. Drawn as the headers look on the board: 3.3 V (pins 6 and 12) and GND (pins 5 and 11) at the left, pin 1 at the right end of the upper row. The board view in the app shows them the same way (section 2.9):

```
          Pmod JA                          Pmod JB
 pin    6    5    4    3    2    1       6    5    4    3    2    1
      3V3  GND  CH3  CH2  CH1  CH0     3V3  GND  CH11 CH10 CH9  CH8
      3V3  GND  CH7  CH6  CH5  CH4     3V3  GND  CH15 CH14 CH13 CH12
 pin   12   11   10    9    8    7      12   11   10    9    8    7
```

Channel	Pmod pin	FPGA pin		Channel	Pmod pin	FPGA pin
0	JA1	Y18		8	JB1	W14
1	JA2	Y19		9	JB2	Y14
2	JA3	Y16		10	JB3	T11
3	JA4	Y17		11	JB4	T10
4	JA7	U18		12	JB7	V16
5	JA8	U19		13	JB8	W16
6	JA9	W18		14	JB9	V12
7	JA10	W19		15	JB10	W13

The outputs are 3.3 V CMOS (8 mA, fast edges). All channels change on the same clock edge. A channel that is not **enabled** is not driven (high impedance), and so are all channels after power-up and after every start of Forth. JA and JB are the board's fast Pmod headers: their pins have no series resistors, so **an output is not**

protected against a short. Do not connect an output to GND, 3.3 V or another output. For edges of a few ns, keep the wires short and use the GND pin of the same header.

Words. Durations are double numbers in ticks (low cell, then high cell), which the first three words make from times:

Word	Stack	Description
`ns`	`( n -- d )`	n nanoseconds in ticks, rounded down to 5 ns: `100 ns` = 20 ticks.
`us`	`( n -- d )`	n microseconds: n × 200 ticks.
`msec`	`( n -- d )`	n milliseconds: n × 200,000 ticks; n up to 21474. Not `ms`: that word waits (section 5.10).
`pg-at`	`( step -- )`	The step the next step word stores, 0..4095.
`pg-step`	`( levels d -- )`	Store a step: the levels (channel n = bit n) for d ticks, then the next step.
`pg-loop`	`( levels d target count -- )`	Store the last step of a repeat: after it, go back to step *target* until the steps *target*..this one have run *count* times in all; 0 = forever.
`pg-jump`	`( levels d target -- )`	Store a step that continues at step *target* after it.
`pg-wait`	`( levels d -- )`	Store a wait step: it shows its levels, waits for the trigger, then holds them for d ticks.
`pg-end`	`( levels -- )`	Store the end: the generator stops with these levels on the pins.
`pg-start`	`( -- )`	Play from step 0 (also while playing: start again).
`pg-stop`	`( -- )`	Stop; the enabled channels go to their idle levels.
`pg-trig`	`( -- )`	The software trigger: a waiting step goes on.
`pg-enable`	`( mask -- )`	Which channels the generator drives (bit n = channel n).
`pg-idle`	`( levels -- )`	The levels while stopped.
`pg-trigger`	`( source edge -- )`	What wait steps wait for. Source: a shield pin 0..41, 48..51 for BTN0..BTN3, 63 for the software trigger only. Edge: 1 rising, 2 falling, 3 both. `pg-trig` works with any source.
`pg-status`	`( -- u )`	The status bits, below.
`pg-pc`	`( -- step )`	The step playing now.
`pg-out@`	`( -- levels )`	The levels on the channels now.
`pg-sync`	`( -- )`	Wait until the generator has taken the last command or setting (about 0.1 µs). The words above that command or set it do this themselves.

The step words refuse to store while a pattern plays: `pattern running`, then the line is abandoned. `pg-stop` first. The generator keeps its steps until power-off, also over a restart of Forth (which stops it and makes all channels undriven).

Repeats. `pg-loop` marks the **last** step of a repeated block; the block runs from *target* to it. A repeat needs no "begin" step, and repeats may **nest up to four deep**, as long as they end on different steps. A fifth nested repeat does not repeat; status bit 5 shows it.

```
\ a 1 MHz clock on channel 0, a marker on channel 1 every 1000 cycles
pg-stop  3 pg-enable  0 pg-idle  0 pg-at
1 500 ns pg-step              \ 0: clock high
0 500 ns 0 999 pg-loop        \ 1: clock low; steps 0-1 play 999 times
3 500 ns pg-step              \ 2: clock and marker high
0 500 ns 0 pg-jump            \ 3: low; back to step 0, forever
pg-start
```

```
\ BTN0 starts a burst of 8 pulses on channel 0 inside a frame on channel 1
pg-stop  3 pg-enable  0 pg-idle  48 1 pg-trigger  0 pg-at
0 1 us pg-wait                \ 0: wait for BTN0, then 1 us low
3 500 ns pg-step              \ 1: clock and frame high
2 500 ns 1 8 pg-loop          \ 2: clock low; steps 1-2 play 8 times
0 2 us 0 pg-jump              \ 3: 2 us low, then wait again
pg-start
```

Status. `pg-status` returns:

Bit	Value	Meaning
0	1	playing (also while waiting)
1	2	waiting for the trigger
2	4	ended (at a `pg-end` step)
3	8	a step word was refused because a pattern played (cleared by `pg-start`, `pg-stop`)
4	16	a step ran late (below; cleared by `pg-start`)
5	32	repeats nested more than four deep (cleared by `pg-start`, `pg-stop`)
6	64	busy: a command has not reached the generator yet

Timing. Every step lasts exactly its duration, also many 5 ns steps in a row. The generator reads its steps ahead into a 16-step buffer, one step per tick, but each jump or repeat costs it one tick, which it makes up during longer steps. So one rule applies to anything that repeats for long: **one pass must last at least one tick more than it has steps**. A repeat of two 5 ns steps cannot keep up; one 5 ns step and one 10 ns step can. If the generator falls behind, the step playing is stretched by a tick (its levels stay right) and status bit 4 is set. The app warns about such repeats before loading.

`pg-start` plays the first step about 0.1 µs later; until then the pins show the idle levels. A trigger edge is seen 10 to 15 ns after it happens (the input is synchronized to the 200 MHz clock), then the wait step's duration runs. Edges that come while no step is waiting are ignored. With nothing connected, shield inputs read 1 (pull-ups), so a pin trigger on the rising edge needs a signal.

Checking it without an instrument. `pg-out@`, `pg-pc` and the board view (section 2.9) show slow patterns live. The millisecond timer measures long ones:

```
: run-time ( -- ms ) ms@ pg-start begin pg-status 4 and until ms@ swap - ;
pg-stop 0 pg-at  1 1000 msec pg-step  0 pg-end  run-time .   \ 1000
```

Registers. At `$80C0`; the words above use them. Details in PatternGenerator.md.

Address	Read	Write
`$80C0`	status	bit 0 start, bit 1 stop, bit 2 software trigger

Address	Read	Write
`$80C2`	step playing	where the next step is stored
`$80C4`	levels now	the step's levels
`$80C6`, `$80C8`		the step's duration, low and high 16 bits
`$80CA`		operation (bits 15..12: 0 step, 1 repeat, 2 jump, 3 wait, 4 end) and target step (bits 11..0)
`$80CC`		the repeat count; writing it stores the step and moves to the next
`$80CE`	enable mask	enable mask
`$80D0`	idle levels	idle levels
`$80D2`	trigger	trigger: bits 5..0 source, bits 9..8 edge
`$80D4`	where the next step goes	–

5.15 Defining words

Word	Stack	Description
`: name`	`( -- )`	Start a new definition called `name` and switch to compiling.
`;`	`( -- )`	End the definition, make it findable, switch back to interpreting.
`variable name`	`( -- )`	Create a variable (one cell, initially 0). `name` then pushes its address.
`constant name`	`( x -- )`	Create a constant. `name` then pushes x.
`create name`	`( -- )`	Create a word that pushes the address of the data space following it; reserve that space with `allot`, `,` or `c,`.
`immediate`	`( -- )`	Make the newest word immediate: it runs during compilation instead of being compiled.

```
variable count
10 constant ten
create buffer 20 allot
: bump 1 count +! ;
```

The name of a definition can't be found until `;`. So a word can be redefined in terms of its old version:

```
: sq dup * ;
: sq sq 1+ ;      \ the inner sq is the old one
5 sq . 26  ok
```

Use `recurse` to call the word being defined.

5.16 Control structures

These are immediate and work **only inside colon definitions**.

Structure	Description
`f if ... then`	Run the code if f is true.
`f if ... else ... then`	Choose between two branches.
`begin ... f until`	Repeat until f is true (runs at least once).
`begin ... f while ... repeat`	Repeat while f is true, test in the middle.
`begin ... again`	Repeat forever (leave with `exit`, or Ctrl-R from the terminal).
`limit start do ... loop`	Counted loop: the index runs from start to limit − 1.
`limit start do ... n +loop`	Like `loop`, but adds n to the index; ends when the index is ≥ limit. Counts upward only.
`exit`	Return from the current word immediately.
`recurse`	Call the word being defined.

Words for use inside `do` loops:

Word	Stack	Description
`i`	`( -- n )`	Index of the innermost loop.
`j`	`( -- n )`	Index of the next outer loop (in the same definition).
`unloop`	`( -- )`	Remove the loop parameters from the return stack. Required before `exit` inside a loop.

`do` always runs the body at least once, and the index runs up to the limit with wrap-around. So `0 0 do ... loop` runs 65536 times. There is no `?do` and no `leave`.

```
: fact ( n -- n! ) dup 1 > if dup 1- recurse * then ;
7 fact . 5040  ok
: evens 10 0 do i . 2 +loop ;
evens 0 2 4 6 8  ok
: table 3 1 do 3 1 do j i * . loop loop ;
table 1 2 2 4  ok
: find3 10 0 do i 3 = if ." found" unloop exit then loop ." none" ;
```

5.17 Compiler and execution

Word	Stack	Description
`' name`	`( -- xt )`	Execution token of `name`; **0 if not found** (standard Forth signals an error).
`['] name`	`( -- )`	Compile only: compile `name`'s xt as a literal.
`execute`	`( xt -- )`	Run the word with this execution token. `0 execute` restarts Forth through `cold`.
`[`	`( -- )`	Switch to interpreting inside a definition (immediate).
`]`	`( -- )`	Switch back to compiling.
`literal`	`( x -- )`	Compile x as a number (immediate). Use it with `[` and `]` to compute values at compile time.

Word	Stack	Description
`compile,`	`( xt -- )`	Compile a call to xt.
`state`	`( -- a )`	Variable: −1 while compiling, 0 while interpreting.

```
: five [ 2 3 + ] literal ;  ok
five . 5  ok
' five execute . 5  ok
: shout ." loud" ; immediate
: quiet shout ;  loud ok       \ shout ran while quiet was compiled
```

5.18 Comments

Word	Description
`( text)`	Ignore everything up to the closing `)`. Commonly used for stack comments.
`\ text`	Ignore the rest of the line.

Both work at the terminal and inside definitions. Like every Forth word, each needs a space after it.

5.19 Interpreter and system

Word	Stack	Description
`words`	`( -- )`	List all words, newest first (internal words included).
`quit`	`( -- )`	Return to the interpreter loop (the data stack is kept).
`abort`	`( -- )`	Clear the data stack and return to the interpreter loop.
`cold`	`( -- )`	Restart: decimal base, banner, LD0 on, interpreter loop. Your definitions stay.

5.20 Variables used by the system

Word	Content
`dp`	Dictionary pointer: the value `here` returns.
`latest`	Header address of the newest word.
`state`	Compile state (section 5.17).
`base`	Number base (section 5.9).
`tib`	Terminal input buffer (82 bytes).
`#tib`	Number of characters in the current input line.
`>in`	Parse position in `tib`.
`'a`, `'n`	Address and length of the name being searched by `find-name`.
`sbuf`, `sn`	Buffer and count used by `.s`.

Word	Content
`ssnap`, `sdepth`	The copy of the data stack `snap` makes before each line (20 cells, bottom first) and its depth, -1 while copying; the ARM reads them for the app.

5.21 Internal words

These words are visible in `words` and can be used, but they mainly exist for the kernel.

Loop and string runtime: these are compiled by `loop`, `+loop`, `."` and `.`.

Word	Stack	Description
`(loop)`	`( -- f )`	Increment the loop index; true when it reached the limit.
`(+loop)`	`( n -- f )`	Add n to the loop index; true when it is ≥ the limit.
`(.")`	`( -- )`	Print the counted string stored after the call and continue behind it.
`(u.)`	`( u -- )`	Print u without a trailing space.
`digit`	`( u -- c )`	Convert 0..35 to the character `0`..`9`, `A`..`Z`.

Parsing the current input line:

Word	Stack	Description
`parse-name`	`( -- a u )`	Skip spaces, return the next space-delimited word (u = 0 at end of line).
`parse`	`( c -- a u )`	Return the text up to character c and skip past c.
`more?`	`( -- f )`	True if unparsed characters remain on the line.
`in-char`	`( -- c )`	The character at the parse position.
`/string`	`( a u n -- a+n u-n )`	Remove n characters from the front of a string.
`upper`	`( c -- c' )`	Convert a lower-case letter to upper case.

Dictionary search:

Word	Stack	Description
`find-name`	`( a u -- h | 0 )`	Find a visible word by name (case-insensitive), newest first.
`match?`	`( h -- f )`	True if header h is visible and matches the name in `'a`/`'n`.
`same?`	`( a1 a2 u -- f )`	Compare two strings of length u, case-insensitive.
`name>count`	`( h -- a u )`	Name string of a header.
`name>xt`	`( h -- xt )`	Execution token of a header.

Numbers:

Word	Stack	Description
`number?`	`( a u -- n f )`	Convert a string with optional `-` and `$` prefix; f is true on success.
`>number`	`( u1 a1 n1 -- u2 a2 n2 )`	Accumulate digits into u1 until a non-digit; returns the rest of the string.

Word	Stack	Description
`digit?`	`( c -- n f )`	Value of a digit character and whether it is valid in `base`.

Compiler:

Word	Stack	Description
`header`	`( a u -- )`	Lay down a header with this name and make it `latest`.
`flags!`	`( mask -- )`	OR mask into the count byte of `latest`.
`hide`	`( -- )`	Mark `latest` hidden (used by `:`).
`reveal`	`( -- )`	Clear the hidden flag of `latest` (used by `;`).
`compile-call`	`( xt -- )`	Compile a call instruction to xt.
`inline,`	`( xt -- )`	Copy xt's instructions up to its return into the definition.
`>resolve`	`( a -- )`	Patch the jump at a to point to `here` (used by `then`, `else`, `repeat`).

Interpreter:

Word	Stack	Description
`snap`	`( -- )`	Copy the data stack into `ssnap` and its depth into `sdepth`, leaving the stack as it was; `quit` runs it before each line, for the app's stack view (section 2.7).
`accept-line`	`( -- )`	Read one line from the terminal into `tib`, with echo and backspace.
`interpret`	`( -- )`	Interpret or compile the rest of the line.
`interpret-word`	`( a u -- )`	Handle one word: execute, compile, inline, or convert as a number.
`?stack`	`( -- )`	Check for stack underflow or overflow (section 3.4).

6. Examples

Each example can be typed at the terminal line by line. The ones that loop until a key is pressed stop on any key. An input line holds 80 characters, so longer definitions are split over several lines or into smaller words, as in the button counter.

6.1 LEDs and buttons

Count in binary on the LEDs

```
: count16 16 0 do i led! 250 ms loop 0 led! ;
count16
```

Buttons light their LEDs

```
: mirror begin btn@ led! key? until key drop 0 led! ;
mirror
```

Running light, speed from the switches

```
: delay sw@ 1+ 100 * ms ;
: runner begin 4 0 do 1 i lshift led! delay loop key? until key drop 0 led! ;
runner
```

SW0 and SW1 select 100, 200, 300 or 400 ms per step.

Colour cycle on both RGB LEDs

```
: colours 8 1 do i dup 3 lshift or rgb! 400 ms loop 0 rgb! ;
colours
```

Reaction timer

```
: wait-btn begin btn@ until ;
: wait-free begin btn@ 0= until ;
: react
  ." wait for the LEDs..." cr
  2000 ms 15 led! ms@
  wait-btn ms@ swap - 0 led!
  ." reaction time " . ." ms" cr
  wait-free ;
react
```

Button counter

```
variable presses
: btn0? btn@ 1 and ;
: count-press 1 presses +! presses @ led! begin btn0? 0= until ;
: counter
  0 presses !
  begin btn0? if count-press then key? until key drop
  ." BTN0 pressed " presses ? ." times" cr ;
counter
```

6.2 Shield connector

These need no extra hardware unless stated. Pin numbers are the shield labels.

Self-test without wires: drive A0 as a digital pin and measure it on the analog channel of the same pin

```
14 output  14 high  10 ms  0 mv@ .
14 low  10 ms  0 mv@ .  14 input
```

The first reading is about 3250 mV, the second a few mV. Driving pins 14..19 in turn lights up channels 0..5, one at a time:

```
: all6 6 0 do i mv@ . loop ;
: drive dup output dup high 50 ms all6 input ;
14 drive                    \ prints 3249 5 5 5 9 5
```

Check the chip's supplies and temperature

```
: supplies 1 supply@ . 2 supply@ . 6 supply@ . 13 supply@ . 14 supply@ . ;
supplies .temp
```

Temperature monitor: one line per second

```
: thermo begin .temp cr 1000 ms key? until key drop ;
thermo
```

Blink an LED on IO13 (LED with a 330 Ω resistor from IO13 to GND)

```
: blink13 13 output begin 13 toggle 250 ms key? until key drop 13 low ;
blink13
```

A button on IO2 (push button from IO2 to GND, no resistor; the pull-up holds the pin high until the button is pressed). LD0 shows the button.

```
: watch2 begin 2 pin@ 0= 1 and led! key? until key drop 0 led! ;
watch2
```

Loopback: connect IO0 to IO1 with a wire

```
0 output  0 high  1 pin@ .  0 low  1 pin@ .  0 input
```

prints `-1 0`.

Voltmeter on A0: prints the voltage every half second

```
: volts begin 0 mv@ . ." mV" cr 500 ms key? until key drop ;
volts
```

Bar graph: a potentiometer (ends to 3.3 V and GND, wiper to A0) lights 0 to 4 LEDs

```
: level ( -- n ) 0 adc@ 820 / 1 swap lshift 1- ;
: bar begin level led! 50 ms key? until key drop 0 led! ;
bar
```

6.3 Pattern generator

These need no extra hardware: the board view (tick **Board**) shows the Pmod channels.

A slow counter on channels 0..3, one count every quarter second, visible in the board view:

```
pg-stop  $F pg-enable  0 pg-at
: counts 16 0 do i 250 msec pg-step loop ;
counts  15 pg-at  15 250 msec 0 pg-jump  pg-start
```

Step 15 is stored again as a jump back to step 0. `pg-pc .` shows the step playing.

A square wave of 10 MHz on channel 0 (50 ns high, 50 ns low):

```
pg-stop  1 pg-enable  0 pg-at
1 50 ns pg-step  0 50 ns 0 pg-jump  pg-start
```

Waiting for a button. A press of BTN1 gives a 2 µs pulse on channel 8 (JB1). The generator sees the button directly, without debouncing, so a bouncing contact can give a few pulses:

```
pg-stop  $100 pg-enable  49 1 pg-trigger  0 pg-at
0 1 0 pg-wait  $100 2 us pg-step  0 1 0 0 pg-jump  pg-start
pg-status .                \ 3: playing and waiting
```

The example patterns in `examples/patterns` open in the app's *Pattern* tab (section 2.6): `counter`, `burst` and `clock-marker`.

7. Differences from standard Forth

HF16 Forth follows the usual Forth conventions (ANS Forth / Forth 2012) where it has the same words, with these differences:

Behaviour

- `'` returns 0 for an unknown word instead of signalling an error, and `0 execute` restarts Forth.
- `/`, `mod` and `/mod` use symmetric division, rounding toward zero (the standard allows this or floored division).
- `+loop` counts upward only.
- No warning when a word is redefined.
- Number overflow is not detected; values wrap around at 16 bits.
- The data stack holds at most 20 items at the terminal (section 3.4); the return stack has 32 entries.
- Compile-only words (`if`, `do`, `."`, `[']`, ...) are not checked. Typed outside a definition, they write stray code into the dictionary and leave junk on the stack.

Not implemented

- Double-cell numbers and most of their words (`d+`, `m*`, `*/mod`, `#`, `<# #>`, ...); only `um*`, `um/mod` and `*/` exist
- `char` and `[char]` (`[char]` exists only in the cross-compiler), `s"`, `accept`, `evaluate`, `source`
- `?do`, `leave`, `case`, `does>`, `postpone`, `defer`, `marker`, `forget`
- `pick`, `roll`, `2swap`, `2over`, `fill`, `move`, `erase`, `see`, `dump`

Most of these can be added as definitions in `forth/kernel.fs` (section 8).

8. Changing the kernel

The kernel lives in `forth/kernel.fs`, written in Forth. `tools/hf16.py` cross-compiles it on the PC. The cross-compiler understands this subset:

- **Outside definitions:** numbers, `:`, `:inline`, `variable`, `constant`, `create`, `allot`, `,`, `c,`, `immediate`, and comments.
- **Inside definitions:** primitives (always inline), calls to other words (including words defined further down), numbers, `if else then begin until again while repeat do loop +loop i unloop exit recurse ." [char] [']`, and `[op] name`, which compiles the opcode of primitive `name` as a number (used by the target compiler).
- `:inline name ... ;` makes a word that the target compiler copies inline.

Workflow:

1. Edit `forth/kernel.fs`.
2. Test in the instruction-set simulator. It runs the kernel and feeds it input lines: `powershell python tools/hf16.py run forth/kernel.fs -f sim/smoke.txt`
3. Optionally run the same input on the Verilog CPU (Vivado simulator, a few minutes): `powershell .\sim\run_sim.ps1`
4. Rebuild the ARM program (which embeds the image) and load it. No new bitstream is needed: `powershell xsct sw/build_sw.tcl xsct sw/run.tcl` For the power-up, build `BOOT.bin` and write it to flash (see the README).

`tools/hf16.py build` also writes `build/forth.map`, the address of every word.

9. Troubleshooting

Symptom	Cause and remedy
No banner, nothing happens	The banner came before you connected: press Ctrl-R (or *Restart* in the app). Check the port and 115200 baud. If LD0 is dark, ZYNQ-FORTH is not running: check JP4 (QSPI) and the flash (chapter 1).
The app says the board does not answer	Another program has the port, the wrong port is chosen, or the board runs other firmware.
`VERIFY FAILED` at the start	The kernel image didn't read back correctly; the CPU stays in reset. Switch the board off and on.
A PWM change does not show at once	It waits for the end of the running cycle, so a channel at 1 Hz takes up to a second (section 5.13).
The board hangs at power-up, or misbehaves from the start	The saved source does it: switch on with BTN3 held to skip it (section 4.7).
A0..A5 read about 0 mV after loading over JTAG	JP4 was on QSPI, so the FSBL from flash ran first. Set JP4 to JTAG, power-cycle and load again.
Forth stops responding	A word is in an endless loop or waiting for `key`. Press any key, or Ctrl-R (*Restart* in the app).
Your words are gone	A restart or a power cycle reloads the original kernel. Keep useful definitions in a file and run it, save it as the board's source (section 2.8), or add them to `kernel.fs`.
Pasting a long text into a terminal program loses characters	The ARM queues up to 64 KB for Forth, so this only happens with very large pastes; the app's *Run all* sends large texts in pieces.
Numbers look wrong, or every number gives `?`	Check `base`: after `hex` or `2 base !`, input and output both use that base, and an error doesn't reset it. `decimal` switches back.
A long line gives an error near its end	Only 80 characters fit on a line; the rest is dropped, possibly in the middle of a word. Split the line.
A shield output doesn't change	The pin is still an input: `n output` first. Check the pin number against the board label (section 5.11).
An analog reading is near 3300 mV on A0..A5 without a signal	The same pin is a digital output driving high (`14 input` … `19 input`).
An analog reading on A6..A11 stays at 0	The − pin is above the + pin, or the − pin is not connected to GND.
`pattern running` when storing pattern steps	The generator plays: `pg-stop` first (section 5.14). The app's *Load & start* does it for you.
The pattern plays but the Pmod pins stay undriven	The channels are not enabled: `$FFFF pg-enable` (or tick them in the app). Every start of Forth disables them.
`pg-status` shows bit 4 (late), the board view *late*	A repeat or jump loop with steps too short for the generator: one pass needs at least one tick more than it has steps (section 5.14, *Timing*).
A wait step never goes on	The trigger source or edge is wrong, or the pin never changes (open shield inputs read 1). `pg-trig` goes on by hand.

Symptom	Cause and remedy
The board view says *firmware 2.2 needed*	The board runs older firmware without the pattern generator: flash the current `BOOT.bin` (chapter 1).